



Rapid metagenomic sequencing for surveillance of bacterial, fungal and viral pathogens using SQK-RPB114.24

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This is a rapid method to perform metagenomic sequencing for identification of bacterial, fungal and viral pathogens.

The protocol:

- Consists of a two-in-one method that splits into bacterial/fungal sample preparation and viral/acellular sample preparation
- Provides a recommended host depletion and extraction method for respiratory samples
- Uses a shotgun approach to RT-PCR
- Involves tagmentation, barcoding and PCR amplification
- Is compatible with R10.4.1 flow cells

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1. Overview of protocol



This is an Early Access product.

For more information about our Early Access programmes, please see [this article on product release phases](#).

Please ensure you always use the most recent version of the protocol.

Introduction to the rapid metagenomic sequencing protocol

This protocol outlines a method to perform agnostic metagenomic sequencing from extracted nucleic acid.

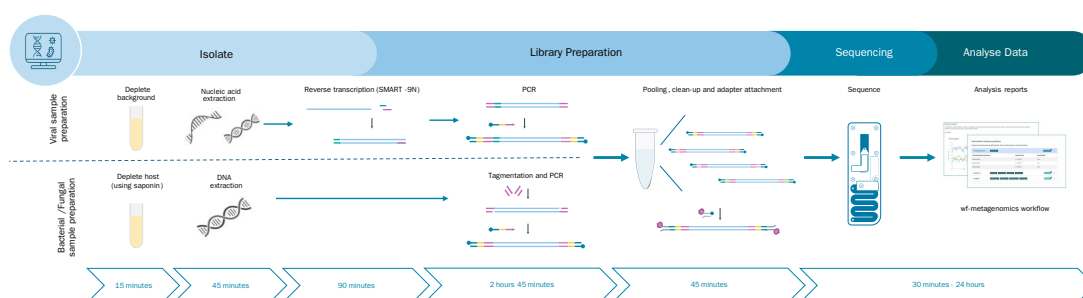
The method offers two options for sample preparation depending on your target sample and input:

- The DNA-only bacterial/fungal sample preparation utilises SQK-RPB114.24 reagents to tagment all DNA in the extract for amplification and sequencing.
- The DNA/RNA viral sample preparation has been optimised by Oxford Nanopore Technologies, and is derived from a method established by [Josh Quick and Ingra M. Claro](#) which utilises a shotgun approach using 9N primers to randomly reverse transcribe RNA and subsequently PCR-amplify DNA/RNA present in a sample.

Other areas of note:

- The performance of this method is reliant on sample type and was optimised for respiratory samples.
- While effort is made to reduce host background, sample types that are less complex/have less nucleic acid background are likely to perform better.

This protocol uses the Rapid PCR Barcoding Kit V14 (SQK-RPB114.24), which allows the potential to use up to 24 barcodes in one sequencing experiment.



For more information and example data from using this protocol please see our [Rapid metagenomic sequencing for surveillance of bacterial, fungal and viral pathogens using SQK-RPB114.24 know-how document](#).

Steps in the sequencing workflow:

Prepare for your experiment

You will need to:

- Ensure you have your sequencing kit, the correct equipment and third-party reagents.
- Extract your DNA/RNA if not using the respiratory sample processing workflow in this protocol.
- Download the software for acquiring and analysing your data.
- Check your flow cell to ensure it has enough pores for a good sequencing run.

Host depletion and extraction

- If using the recommended respiratory sample host depletion and extraction, ensure you have the correct third-party reagents.
- Ensure you have fresh samples to use (not recommended for frozen/deactivated samples).

Library preparation

The Table below is an overview of the steps required in the library preparation, including timings and optional stopping points.

Step	Process	Time	Stop option
Viral samples: reverse transcription	Reverse transcribe your RNA samples with the RLB RT 9N primer mix and the TSOmG template-switching oligo.	~90 minutes	
Bacterial/fungal samples: DNA tagmentation	Tagment your DNA using the Fragmentation Mix from the sequencing kit.	10 minutes	
PCR amplification	PCR your sample using the barcoded primer supplied in the sequencing kit.	180 minutes	<i>Optional:</i> The PCR amplification can be performed and left at the hold temperature overnight.
Sample quantification and pooling	Quantify your barcoded PCR samples and pool them in equimolar ratios.	15 minutes	
Clean-up and quantification	Perform a purification on your pooled samples and quantify.	15 minutes	4°C overnight
Rapid adapter attachment and clean-up	Attach the sequencing adapters to the DNA ends.	5 minutes	We strongly recommend sequencing your library as soon as it is adapted.
Priming and loading the flow cell	Prime the flow cell and load the prepared library for sequencing.	10 minutes	

Sequencing and analysis

You will need to:

- Start a sequencing run using the MinKNOW software, which will collect raw data from the device and convert it into basecalled reads.
- Analyse your data using the [wf-metagenomics workflow](#) available through EPI2ME or the command line.



Compatibility of this protocol

This protocol should only be used in combination with:

- Rapid PCR Barcoding Kit (SQK-RPB114.24)
- R10.4.1 flow cells (FLO-MIN114)
- Flow Cell Wash Kit (EXP-WSH004)
- Rapid Adapter Auxiliary V14 (EXP-RAA114)
- Sequencing Auxiliary Vials V14 (EXP-AUX003)
- Flow Cell Priming Kit V14 (EXP-FLP004)
- MinION Mk1D - [MinION Mk1D IT requirements document](#)
- GridION - [GridION IT requirements document](#)

2. Equipment and consumables

Materials	(FOR SAMPLE PREPARATION) Bacterial/fungal or viral/acellular sample inputs (see details below)
	(FOR LIBRARY PREPARATION) Extracted nucleic acid
	Rapid PCR Barcoding Kit 24 V14 (SQK-RPB114.24)

Consumables	MinION/GridION Flow Cell
	Sputasol (Fisher, 11943262)
	RT-PCR Grade Water (Fisher, 10289774)
	Phosphate Buffer Saline (PBS) (Fisher, 11593377)
	2X HL-SAN Buffer (4.5 M NaCl and 100 mM MgCl ₂) formulated using the two reagents below:
	Invitrogen 5M NaCl (Fisher, 10255984)
	Invitrogen 1M MgCl ₂ (Fisher, 10418464)
	HL-SAN Triton Free DNase (ArticZymes, 70911-202)
	Saponin solution (0.2% saponin in PBS) formulated using the reagent below:
	Saponin (Sigma-Aldrich, 47036-50G-F)
	Matrix Lysing E tubes (Fisher, 11452420)
	MagMAX Viral/Pathogen Nucleic Acid Isolation Kit (Fisher, 16346582)
	12 µM RLB RT 9N primer (IDT - Sequence: 5'-TTTTTCGTGCGCCGCTTCAACNNNNNNNNN-3')
	12 µM TSOmG template-switching oligo (IDT - Sequence: 5'-GCTAATCATTGCTTTTCGTGCGCCGCTTCAACATmGmGmG-3')
	10 mM dNTPs (Fisher, 10610851)
	Maxima H Minus Reverse Transcriptase (Fisher, 13233159)
	RNaseOUT (Fisher, 10777019)
	LongAmp Hot Start Taq 2X Master Mix (NEB, M0533)
	Thermolabile Exonuclease I (NEB, M0568)
	Agencourt AMPure XP beads (Beckman Coulter™, A63881)
	Nuclease-free water (e.g. Thermo Scientific, AM9937)

Bovine Serum Albumin (BSA) (50 mg/ml) (e.g. Invitrogen™ UltraPure™ BSA 50 mg/ml, AM2616)
1.5 ml Eppendorf DNA LoBind tubes
0.2 ml thin-walled PCR tubes
Qubit™ dsDNA HS Assay Kit (ThermoFisher, Q32851)
Qubit™ Assay Tubes (Invitrogen, Q32856)

Equipment

MinION or GridION device
MinION/GridION Flow Cell Light Shield
Benchtop microcentrifuge (max speed 15,000 rpm)
Microfuge
Timer
Vortex mixer
Thermomixer set at 37°C
Thermomixer set at 65°C
Magnetic separation rack
Bead-beater (e.g. FastPrep-24™ Classic bead beating grinder and lysis system)
Thermal cycler
96-well PCR block
Ice bucket with ice
Hula mixer (gentle rotator mixer)
Qubit fluorometer (or equivalent)
P1000 pipette and tips
P200 pipette and tips
P100 pipette and tips
P20 pipette and tips
P10 pipette and tips
P2 pipette and tips

Optional equipment

Suitable biosafety equipment for your sample(s) (e.g. Microbiological Safety Cabinet II contamination level 3 or equivalent)



Ensure pathogenic samples are handled in appropriate biosafety conditions.

Please adhere to the correct health and safety practices in accordance to your laboratory standards and local rules and regulations.

Note: We recommend as a minimum the use of a Class II Biological Safety Cabinet. Please consider that some organisms can survive bead-beating steps during extraction. Ensure you are taking the necessary precautions during your sample and library preparation.

For this protocol you will need the following sample input(s):

For sample preparation:

- ≥ 250 μ l of Sputum/Endotracheal aspirates

OR

- ≥ 250 μ l of Bronchoalveolar lavages/Mini-BALs

OR

- ≥ 500 μ l of VTM/UTM-stored oral/nasal swabs (only suitable for viral samples)

Note: If quantities allow, a sample can be processed for both bacterial/fungal sample preparation and viral sample preparation. Please note, each preparation type will differ in method and should be treated as a separate sample.

For library preparation:

- Extracted nucleic acid from the sample types above.

Input DNA/RNA

How to QC your input

It is important that the input DNA/RNA meets the quantity and quality requirements. Using too little or too much DNA/RNA, or DNA/RNA of poor quality (e.g. highly fragmented or containing RNA or chemical contaminants) can affect your library preparation.

For instructions on how to perform quality control of your DNA/RNA sample, please read the [Input DNA/RNA QC protocol](#).

Chemical contaminants

Depending on how the DNA/RNA is extracted from the raw sample, certain chemical contaminants may remain in the purified DNA/RNA, which can affect library preparation efficiency and sequencing quality. Read more about contaminants on the [Contaminants page](#) of the Community.



Input controls

Negative internal controls

We recommend a negative control is included and processed as a real sample through the entire process to monitor for contamination events.

The negative control can be PBS or a third-party negative sample matrix.

Third-party reagents

We have validated and recommend the use of all the third-party reagents used in this protocol. Alternatives have not been tested by Oxford Nanopore Technologies.

For all third-party reagents, we recommend following the manufacturer's instructions to prepare the reagents for use.

Agencourt AMPure XP beads

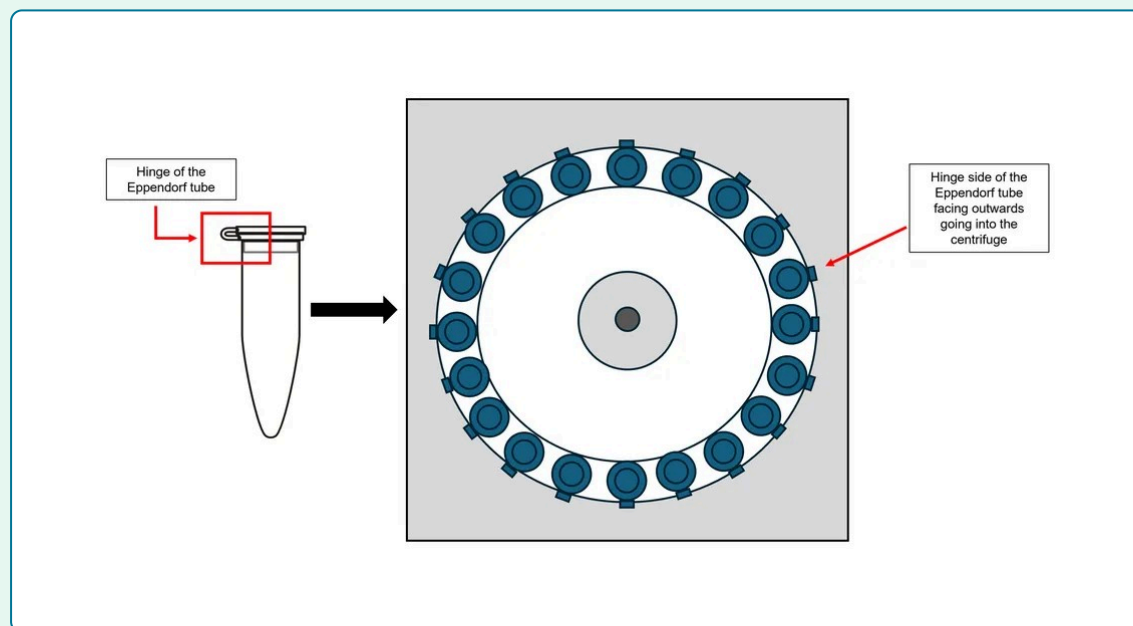
Additional Agencourt AMPure XP beads may be required alongside the AMPure XP Beads (AXP) provided in the sequencing kit for the clean-up step following PCR amplification.



Eppendorf tube orientation in centrifuge

For all centrifugation steps, ensure that tubes are loaded into the centrifuge with the hinge side of the tube facing outwards. This will assist in visual identification of the pellet.

Ensure gentle handling when removing the tubes from the centrifuge to avoid dislodging the pellet.



Check your flow cell

We highly recommend that you check the number of pores in your flow cell prior to starting a sequencing experiment. This should be done within 12 weeks of purchasing your MinION/GridION Flow Cells. Oxford Nanopore Technologies will replace any unused flow cell with fewer than the number of pores listed in the Table below, when the result is reported within two days of performing the flow cell check, and when the storage recommendations have been

followed. To do the flow cell check, please follow the instructions in the [Flow Cell Check document](#).

Flow cell	Minimum number of active pores covered by warranty
MinION/GridION Flow Cell	800



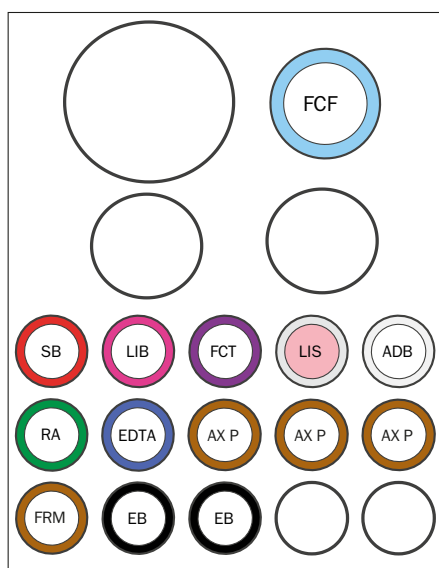
The Rapid Adapter (RA) used in this kit and protocol is not interchangeable with other sequencing adapters.

Rapid PCR Barcoding Kit 24 V14 (SQK-RPB114.24) contents

We are in the process of reformatting the Rapid Barcode Primers provided in this kit into a plate format. Additionally we have lowered the concentration of the Rapid Barcode Primers in the plate format to 1 μ M (down from 10 μ M in the vial format). These changes will reduce plastic waste and facilitates automated applications.

Please ensure you follow the correct instructions in the method for your kit format and Rapid Barcode Primers concentration.

Plate format with Rapid Barcode Primers at 1 μ M



RA:Rapid Adapter
 SB:Sequencing Buffer
 LIB:Library Beads
 LIS:Library Solution
 EB:Elution Buffer
 AX P:AMPure X P Beads

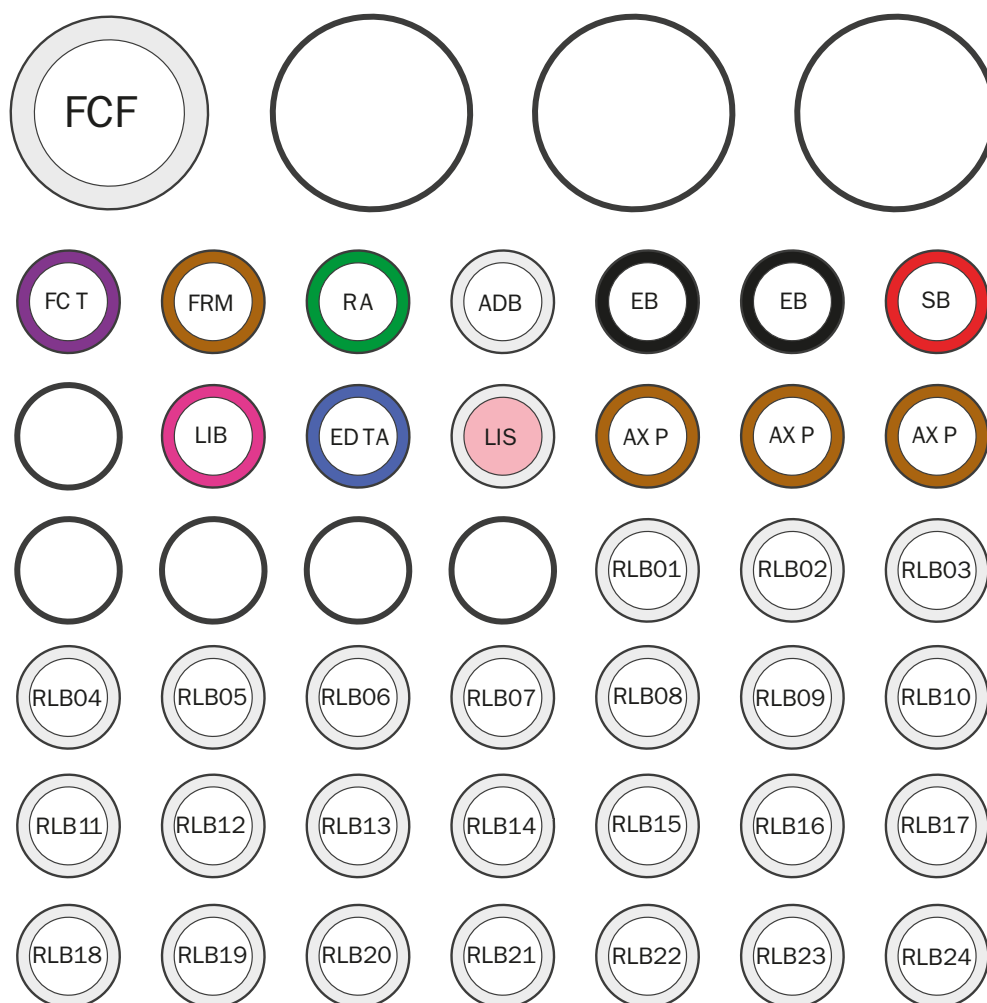
	01	02	03	04	05	06	07	08	09	10	11	12
A	1	9	17	1	9	17	1	9	17	1	9	17
B	2	10	18	2	10	18	2	10	18	2	10	18
C	3	11	19	3	11	19	3	11	19	3	11	19
D	4	12	20	4	12	20	4	12	20	4	12	20
E	5	13	21	5	13	21	5	13	21	5	13	21
F	6	14	22	6	14	22	6	14	22	6	14	22
G	7	15	23	7	15	23	7	15	23	7	15	23
H	8	16	24	8	16	24	8	16	24	8	16	24

FCF:Flow Cell Flush
 FCT:Flow Cell Tether
 EDTA:EDTA
 FRM:Fragmentation Mix
 ADB:Adapter Buffer
 RPB:Rapid Barcode Primers 01-24

Name	Acronym	Cap colour	No. of vials	Fill volume per vial (µl)
Fragmentation Mix	FRM	1	Brown	160
Rapid Adapter	RA	1	Green	15
Adapter Buffer	ADB	1	Clear	100
AMPure XP Beads	AXP	3	Amber	1,200
Elution Buffer	EB	2	Black	500
EDTA	EDTA	1	Blue	700
Sequencing Buffer	SB	1	Red	700
Library Beads	LIB	1	Pink	600
Library Solution	LIS	1	White cap, pink label	600
Flow Cell Flush	FCF	1	Clear cap, light blue label	8,000
Flow Cell Tether	FCT	1	Purple	200
Rapid Barcode Primers (01-24) at 1 µM	RPB	-	2 plates, 3 sets of primer barcodes per plate	15 µl per well

Note: This product contains AMPure XP reagent manufactured by Beckman Coulter, Inc. and can be stored at -20°C with the kit without detriment to reagent stability.

Vial format with Rapid Barcode Primers at 10 µM



RA :Rapid Adapter
 SB :Sequencing Bu ffer
 LIB :Library Beads
 LIS :Library Solution
 EB :Elution Buffer
 AX P :AMPure XP Beads
 FCF :Flow Cell Flush
 FC T :Flow Cell Tether
 EDTA :ED TA
 FRM :Fragmentation Mix
 ADB :Adapter Bu ffer
 RLB01 :Rapid Barcode Primer 1
 RLB02 :Rapid Barcode Primer 2
 RLB03 :Rapid Barcode Primer 3
 RLB04 :Rapid Barcode Primer 4
 RLB05 :Rapid Barcode Primer 5
 RLB06 :Rapid Barcode Primer 6

RLB07 :Rapid Barcode Primer 7
 RLB08 :Rapid Barcode Primer 8
 RLB09 :Rapid Barcode Primer 9
 RLB10 :Rapid Barcode Primer 10
 RLB 11 :Rapid Barcode Primer 11
 RLB12 :Rapid Barcode Primer 12
 RLB13 :Rapid Barcode Primer 13
 RLB14 :Rapid Barcode Primer 14
 RLB15 :Rapid Barcode Primer 15
 RLB16 :Rapid Barcode Primer 16
 RLB17 :Rapid Barcode Primer 17
 RLB18 :Rapid Barcode Primer 18
 RLB19 :Rapid Barcode Primer 19
 RLB20 :Rapid Barcode Primer 20
 RLB21 :Rapid Barcode Primer 21
 RLB22 :Rapid Barcode Primer 22
 RLB23 :Rapid Barcode Primer 23
 RLB24 :Rapid Barcode Primer 24

Name	Acronym	Cap colour	No. of vials	Fill volume per vial (µl)
Fragmentation Mix	FRM	Brown	1	160
Rapid Adapter	RA	Green	1	15
Adapter Buffer	ADB	Clear	1	100
AMPure XP Beads	AXP	Amber	3	1,200
Elution Buffer	EB	Black	2	500
EDTA	EDTA	Blue	1	700
Sequencing Buffer	SB	Red	1	700
Library Beads	LIB	Pink	1	600
Library Solution	LIS	White cap, pink label	1	600
Flow Cell Flush	FCF	Clear	1	8,000
Flow Cell Tether	FCT	Purple	1	200
Rapid Barcode Primer 01-24	RLB01-24	Clear	24 (one per barcode)	15

Note: This product contains AMPure XP reagent manufactured by Beckman Coulter, Inc. and can be stored at -20°C with the kit without detriment to reagent stability.

Rapid Barcode Primers

Component	Sequence
Rapid Barcode Primer 01	AAGAAAGTTGTCGGTGTCTTTGTG
Rapid Barcode Primer 02	TCGATTCCGTTTGTAGTCGTCTGT
Rapid Barcode Primer 03	GAGTCTTGTGTCCCAGTTACCAGG
Rapid Barcode Primer 04	TTCGGATTCTATCGTGTTTCCCTA
Rapid Barcode Primer 05	CTTGTCAGGGTTTGTGTAACTT
Rapid Barcode Primer 06	TTCTCGCAAAGGCAGAAAGTAGTC
Rapid Barcode Primer 07	GTGTTACCGTGGGAATGAATCCTT
Rapid Barcode Primer 08	TTCAGGGAACAAACCAAGTTACGT
Rapid Barcode Primer 09	AACTAGGCACAGCGAGTCTTGTT
Rapid Barcode Primer 10	AAGCGTTGAAACCTTTGTCCTCTC
Rapid Barcode Primer 11	GTTTCATCTATCGGAGGGAATGGA
Rapid Barcode Primer 12	GTTGAGTTACAAAGCACCGATCAG
Rapid Barcode Primer 13	AGAACGACTTCCATACTCGTGTGA
Rapid Barcode Primer 14	AACGAGTCTCTTGGGACCCATAGA
Rapid Barcode Primer 15	AGGTCTACCTCGCTAACACCACTG
Rapid Barcode Primer 16	CGTCAACTGACAGTGGTTCGTACT
Rapid Barcode Primer 17	ACCCTCCAGGAAAGTACCTCTGAT
Rapid Barcode Primer 18	CCAAACCCAACAACCTAGATAGGC

Component	Sequence
Rapid Barcode Primer 19	GTTCTCTCGTGCAGTGTCAAGAGAT
Rapid Barcode Primer 20	TTGCGTCCTGTTACGAGAACTCAT
Rapid Barcode Primer 21	GAGCCTCTCATTGTCCGTTCTCTA
Rapid Barcode Primer 22	ACCACTGCCATGTATCAAAGTACG
Rapid Barcode Primer 23	CTTACTACCCAGTGAACCTCCTCG
Rapid Barcode Primer 24	GCATAGTTCTGCATGATGGGTTAG

3. Host depletion and extraction

Materials

Sputum/Endotracheal aspirates
Bronchoalveolar lavages/Mini-BALs
VTM/UTM-stored oral/nasal swabs (viral-arm only)

Consumables

Sputasol (Fisher, 11943262)
RT-PCR Grade Water (Fisher, 10289774)
Phosphate Buffer Saline (PBS) (Fisher, 11593377)
2X HL-SAN Buffer (4.5 M NaCl and 100 mM MgCl₂) formulated using the two reagents below:
Invitrogen 5M NaCl (Fisher, 10255984)
Invitrogen 1M MgCl₂ (Fisher, 10418464)
HL-SAN Triton Free DNase (ArticZymes, 70911-202)
Saponin solution (0.2% saponin in PBS) formulated using the reagent below:
Saponin (Sigma-Aldrich, 47036-50G-F)
Matrix Lysing E tubes (Fisher, 11452420)
MagMAX Viral/Pathogen Nucleic Acid Isolation Kit (Fisher, 16346582)
1.5 ml Eppendorf DNA LoBind tubes
15 ml Falcon tubes

Equipment

Benchtop microcentrifuge (max speed 15,000 rpm)
Microfuge
Timer
Vortex mixer
Thermomixer set at 37°C
Thermomixer set at 65°C
Magnetic separation rack

Bead-beater (e.g. FastPrep-24™ Classic bead beating grinder and lysis system)
P1000 pipette and tips
P200 pipette and tips
P20 pipette and tips
P10 pipette and tips

Optional equipment

Suitable biosafety equipment for your sample(s) (e.g. Microbiological Safety Cabinet II contamination level 3 or equivalent)



Ensure you are keeping your bacterial/fungal samples separate from your viral samples.

If processing a mix of **bacterial/fungal samples** and **viral samples**, ensure you keep them separate to avoid following the wrong sample preparation method.

Both methods can be followed simultaneously if processing different sample types prior to library preparation. Please ensure you are following the correct method for each sample type.

Note: A sample can be processed for both bacterial/fungal sample preparation and viral sample preparation. Please follow the instructions below, where the supernatant is used for viral sample preparation and the pellet is used for bacterial/fungal sample preparation. Please note, each preparation type will differ in method and should be treated as a separate sample.

1 Ensure you have sufficient sample input. You will need:

- **250 µl** of sputum, endotracheal aspirate or other viscous samples (e.g. mucosal BAL).

OR

- **500 µl** of swab sample in transport media or other liquid samples.



Input controls

Negative internal controls

We recommend a negative control is included and processed as a real sample through the entire process to monitor for contamination events.

The negative control can be PBS or a third-party negative sample matrix.

2 Prepare a working solution of Sputasol in nuclease-free water (1:13.33) as follows:

- In a fresh 15 ml Falcon tube, dispense 750 µl Sputasol.
- Add 9.25 ml nuclease-free water.
- Thoroughly mix by vortexing.

Note: Volumes can be adjusted to meet your experiment requirements.

3 Prepare a 0.2% saponin-PBS solution:

Reagent	Quantity/Volume
Saponin (Sigma-Aldrich, 47036-50G-F)	0.01 g
Phosphate Buffer Saline (PBS)	5,000 µl
Total volume 0.2% saponin-PBS solution	5,000 µl

Note: Quantity/volumes can be adjusted to meet your experiment requirements.

The 0.2% saponin-PBS solution can be stored in the fridge (~4°C) for up to 30 days.

4 Prepare the 2X HLSAN Buffer (4.5 M NaCl and 100 mM MgCl₂) as follows:

Reagent	Volume
Invitrogen 5M NaCl	4,500 µl
Invitrogen 1M MgCl ₂	500 µl
Total	5,000 µl

Note: Volumes can be adjusted to meet your experiment requirements.

The formulated 2X HLSAN Buffer can be stored at room temperature (~22°C) long-term.

5 Ensure your thermomixers are set to 37°C and 65°C.

6 Prepare your sputum, endotracheal (ETT), or any other mucoid sample(s):

Note: Transport Media and BAL samples do not require this unless they are mucoid.

- To each sample, add an equal volume (1:1) of Sputasol working solution.
- Mix by vortexing for 30 seconds.
- Incubate at room temperature until liquefaction (at least 5 minutes).

Tip: If the sample is still viscous or sticky after 5 minutes, repeat the process above. Full liquefaction of sputum is important for good depletion and efficient extraction.

7 For each sample, transfer 500 µl into a separate clean 1.5 ml Eppendorf tube.

8 Centrifuge for 5 minutes at 10,000 x g.

9 Carefully remove tubes from centrifuge without disturbing the solution.



Remember, your sample(s) can be processed for both viral sample preparation and bacterial/fungal sample preparation.

Please follow the instructions outlined below and treat each preparation as a separate sample.

10 Perform sample preparation side-by-side depending on your sample type:

Step	For viral sample preparation	For bacterial/fungal samples
1.	For each sample, transfer 300 µl of the supernatant (by aspirating from the top) to a separate new 1.5 ml Eppendorf tube.	OPTIONAL: the remaining pellet and supernatant from the sample used in the viral sample preparation can be taken forward and processed in the bacterial/fungal sample preparation as seen below
2.		Using a pipette, carefully remove most of the supernatant without disturbing the pellet, leaving enough volume to cover the pellet. Note: Ensure there is enough supernatant left (~50 µl) above the pellet so as to not disturb it.
3.		Thoroughly resuspend the pellet in 200 µl of the prepared 0.2% saponin-PBS solution. Mix by pipetting.
4.	Add the depletion reagent to the sample supernatant: • Add 10 µl of HLSAN enzyme. • Vortex mix for 3 seconds.	Add the depletion reagents to the resuspended pellet as follows: • Add 200 µl of the prepared 2X HLSAN Buffer. • Add 10 µl of HLSAN enzyme. • Vortex mix for 3 seconds.
5.	Incubate the reaction on a thermomixer at 37°C for 10 minutes at 1,000 rpm.	Incubate the reaction on a thermomixer at 37°C for 10 minutes at 1,000 rpm.
6.	To each tube containing the sample supernatant, prepare the following reaction:	

Reagent	Volume per sample (µl)
Sample supernatant in the tube (from previous step)	310
Proteinase K from the MagMAX kit	10
Binding Buffer from the MagMAX kit	520
Magnetic beads from the MagMAX kit	20
Total	860

Step	For viral sample preparation	For bacterial/fungal samples
	Ensure the reagents are mixed by inverting the tube multiple times. Avoid mixing the lysis reagents too vigorously or with any method that would lead to foam (e.g. vortexing). Note: A mastermix of the extraction reagents can be made and added together instead.	
7.		Add 900 µl of PBS to your sample(s) and mix by pipetting.
8.		Centrifuge your sample(s) at 12,000 x g for 3 minutes.
9.		Using a pipette, carefully remove most of the supernatant without disturbing the pellet, leaving enough volume to cover the pellet. Ensure there is enough supernatant left (~50 µl) above the pellet so as to not disturb it.
10.		Resuspend pellet in 500 µl of PBS and mix by pipetting.
11.		For each sample, transfer whole volume of sample into a separate new bead-beating tube (matrix lysing E tubes).

Step	For viral sample preparation	For bacterial/fungal samples
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12.

Insert the bead-beating tubes into the FastPrep-24 Classic bead beating grinder and lysis system and run the program as follows:

Condition	Setting
Speed	6.0 m/s
Time	30 seconds

13.

Remove the bead-beating tubes from the Fast-prep device, and centrifuge for 30 seconds at max speed ($>20,000 \times g$).

14.

For each sample, transfer 300 μ l of the supernatant to a separate new 1.5 ml Eppendorf tube.

Note: Avoid aspirating the beads from the bead-beating tube.

15.

To each tube containing the sample supernatant, prepare the following reaction:

Reagent	Volume per sample (μ l)
Sample supernatant in the tube (from previous step)	300
Proteinase K from the MagMAX kit	10
Binding Buffer from the MagMAX kit	520
Magnetic beads from the MagMAX kit	20
Total	850

Ensure the reagents are mixed by inverting the tube multiple times.

Step	For viral sample preparation	For bacterial/fungal samples
		Avoid mixing the lysis reagents too vigorously or with any method that would lead to foam (e.g. vortexing). Note: A mastermix of the extraction reagents can be made and added together instead.

11 Transfer all sample tubes to a thermomixer and incubate the reactions at 65°C for 5 minutes at 1,000 rpm.

12 Transfer the sample tube(s) to a hulamixer and incubate/mix at room temperature for 5 min.

Hulamixer settings:

- Orbital 15 rpm (10 seconds)
- Reciprocal 70° (15 seconds)
- Vibro/pause 5° (3 seconds)
- 5 minutes total

Note: Tube inversion is crucial during this incubation, as the beads may collect at the bottom of the tube. We highly recommend the use of a hulamixer; however if this is not available, ensure the samples are manually inverted during the incubation.

13 Prepare 2 ml of fresh 80% ethanol per sample in nuclease-free water.

14 Briefly spin down the tube(s) and pellet on a magnetic rack until supernatant is clear and colourless (for at least 5 minutes). Keep the tube on the magnetic rack, and pipette off the supernatant.

Take care not to disturb the pelleted beads.

15 Remove the tube from the magnetic rack and add 1 ml of Wash Buffer from the MagMAX kit. Gently mix by inverting the tube until fully resuspended.

Note: Avoid techniques that will create foam when mixing your tubes. This can negatively impact sample recovery.

- 16** Briefly spin down the tube(s) and pellet on a magnetic rack until supernatant is clear and colourless (for at least 2 minutes). Keep the tube on the magnetic rack, and pipette off the supernatant.
- 17** Remove the tube from the magnetic rack and add 1 ml of 80% ethanol. Gently mix by inverting the tube until fully resuspended.
- 18** Briefly spin down the tube(s) and pellet on a magnetic rack until supernatant is clear and colourless (for at least 2 minutes). Keep the tube on the magnetic rack, and pipette off the supernatant.
- 19** Remove the tube from the magnetic rack and add 500 µl of 80% ethanol. Gently mix by inverting the tube until fully resuspended.
- 20** Briefly spin down the tube(s) and pellet on a magnetic rack until supernatant is clear and colourless (for at least 2 minutes). Keep the tube on the magnetic rack, and pipette off the supernatant.
- 21** Keeping the tube on the magnetic rack, leave the lid open and allow to dry for ~2 minutes, but do not dry the pellet to the point of cracking.
- 22** Remove the tube from the magnetic rack and resuspend the pellet by pipetting in 20 µl of nuclease-free water. Ensure the pellet is fully resuspended by pipette mixing.
- 23** To aid with sample elution, transfer tubes to a thermomixer and incubate at 65°C for 5 minutes at 1,000 rpm.
- 24** Pellet the beads on a magnet until the eluate is clear and colourless for at least 1 minute.
- 25** Remove and retain the eluate (~15 µl) into a clean 1.5 ml Eppendorf DNA LoBind tube.

- Dispose of the pelleted beads



Take your extracted sample forward into the library preparation section of this protocol.

Alternatively, if you are not using your sample immediately, store at -20°C.

4. Reverse transcription and PCR

Materials

For viral sample preparation: 10 µl of extracted nucleic acid from previous step

For bacterial/fungal sample preparation: 3 µl of extracted nucleic acid from previous step

Fragmentation Mix (FRM)

Rapid Barcode Primers (vial format (RLB01-24, at 10 µM) or plate format (RPB01-24, at 1 µM))

Consumables

12 µM RLB RT 9N primer (IDT - Sequence: 5'-TTTTTCGTGCGCCGCTTCAACNNNNNNNNN-3')

12 µM TSOmG template-switching oligo (IDT - Sequence: 5'-GCTAATCATTGCTTTTCGTGCGCCGCTTCAACATmGmGmG-3')

10 mM dNTPs (Fisher, 10610851)

Maxima H Minus Reverse Transcriptase (Fisher, 13233159)

RNaseOUT (Fisher, 10777019)

LongAmp Hot Start Taq 2X Master Mix (NEB, M0533)

Nuclease-free water (e.g. Thermo Scientific, AM9937)

1.5 ml Eppendorf DNA LoBind tubes

0.2 ml thin-walled PCR tubes

Equipment

Hula mixer (gentle rotator mixer)

Microfuge

Thermal cycler

96-well PCR block

Timer

Ice bucket with ice

P1000 pipette and tips

P200 pipette and tips

P100 pipette and tips

P20 pipette and tips

P10 pipette and tips

P2 pipette and tips



Ensure you are following the correct steps in the method for your sample type.

The method differs for bacterial/fungal sample processing and for viral sample processing. Please ensure you keep track of your samples to ensure you are following the correct method.

Note: If a sample has been processed for both bacterial/fungal sample preparation and viral sample preparation, you will have two "separate" samples to process and barcode in library preparation.

1 Thaw kit components at room temperature, then spin down briefly using a microfuge and mix as indicated by the Table below. Then place on ice:

Reagents	1. Thaw at room temperature	2. Briefly spin down	3. Mix well by pipetting
Fragmentation Mix (FRM)	Not frozen	✓	✓
Rapid Barcode Primers (vial format (RLB01-24, at 10 µM) or plate format (RPB01-24, at 1 µM))	✓	✓	✓
RLB RT 9N primer (12 µM)	✓	✓	✓
TSOmG template-switching oligo (12 µM)	✓	✓	✓
10 mM dNTP solution	✓	✓	✓
Maxima H Minus Reverse Transcriptase	Not frozen	✓	✓
Maxima H Minus 5x RT Buffer	✓	✓	✓
LongAmp Taq 2X Master Mix	✓	✓	✓
RNaseOUT	Not frozen	✓	✓



The wells of the barcoding plate are intended for single use only. Please ensure your barcode well is sealed before use, and do not reuse the barcode well once pierced/opened.

2 Prepare each of your extracted nucleic acid samples from the previous step in a separate clean 0.2 ml tube:

For viral sample preparation:

- Take forward 10 µl of each viral nucleic acid extract.

Note: Quantifying viral samples is not recommended, as these will likely be below the limit of detection for Qubit High Sensitivity or similar fluorescence-based quantification methods. Please proceed with the recommended volume of viral nucleic acid extract.

For bacterial/fungal samples:

- Quantify 1 µl of each bacterial/fungal extraction using Qubit High Sensitivity.
 - **If the extract is more than 1 ng/µl:** dilute to 1 ng/µl using nuclease-free water.
 - **If the extract is less than 1 ng/µl (or unquantifiable):** no dilution is necessary, and extract is used directly.
- Take forward 3 µl of each bacterial/fungal DNA extract.

3 Perform the RT reaction / tagmentation reaction side-by-side depending on your sample type:

Note: If processing both sample types we recommend preparing the viral samples first as they have a longer incubation time. The bacterial/fungal samples can be processed during the viral sample incubation.

Step	For viral sample preparation (RT reaction)	For bacterial/fungal samples (tagmentation)
------	--	---

1. In a clean 0.2 ml thin-walled PCR tube, prepare the following master mix.

Tip: Generate sufficient volume for each viral sample preparation + 1 extra volume for excess:

Reagent	Volume per sample (µl)
RLB RT 9N (12 µM)	1
dNTP mix (10 mM)	1
Total	2

2. Mix the master mix prepared above by vortexing and spin down briefly.
3. Add 2 µl of the prepared mastermix to each of your 10 µl of viral nucleic acid extract.
4. Incubate the reaction at 65°C for 5 minutes, then transfer the samples to ice immediately.
5. Keep your samples on ice for 2 minutes.
6. In a clean 0.2 ml thin-walled PCR tube, prepare the Maxima H master mix.

Tip: Generate sufficient volume for each viral sample preparation + 1 extra volume for excess:

Step	For viral sample preparation (RT reaction)	For bacterial/fungal samples (tagmentation)
------	--	---

Reagent	Volume per sample (µl)
Maxima H(-) buffer (5X)	4
TSOmG (12 µM)	2
RNase OUT	1
Maxima H minus enzyme	1
Total	8

Note: Add the Maxima H Enzyme last. Keep the master mix on ice.

7. ~~Add 8 µl of the Maxima H master mix to each of your sample tubes for a total of 24 µl. The enzymes are not hot-start, and the reaction will start as soon as it comes in contact with your~~

8. ~~Pre-heat a PCR block to 42°C, then add your sample tubes and incubate using the following conditions:~~

During the RT reaction (viral samples), the bacterial/fungal samples can be prepared.

Temperature	Time
42°C	75 min
80°C	5 min
10°C	Hold

- 9.

To each of the 3 µl of bacterial/fungal DNA extract, add 1 µl of FRM, and mix gently by flicking the tube.

- 10.

Pre-heat a PCR block to 30°C, then add your sample tubes and incubate using the following conditions:

Step	For viral sample preparation (RT reaction)	For bacterial/fungal samples (tagmentation)								
		<table><tr><th>Temperature</th><th>Time</th></tr><tr><td>30°C</td><td>2 min</td></tr><tr><td>80°C</td><td>2 min</td></tr><tr><td>10°C</td><td>Hold</td></tr></table>	Temperature	Time	30°C	2 min	80°C	2 min	10°C	Hold
Temperature	Time									
30°C	2 min									
80°C	2 min									
10°C	Hold									



Please ensure you are following the correct method for your SQK-RPB114.24 kit format.

In addition to the kit format change, we have lowered the concentration of the Rapid Barcode Primers in the plate format to 1 μ M (down from 10 μ M in the vial format) to facilitate automation usage and improve kit usability.

Please note, this will mean that the volumes used in the PCR reaction set up will vary depending on the kit format you have available.

4 In a clean 1.5 Eppendorf tube, prepare the following master mix:

Tip: Generate sufficient volume for each sample preparation + 1 extra volume for excess:

Plate format with Rapid Barcode Primers at 1 μ M

Reagent	Volume
LongAmp Hot Start Taq 2X Master Mix	25 μ l
Nuclease-free water	11 μ l
Total volume	36 μl

Mix by pipetting, taking care not to generate bubbles or foam.

Vial format with Rapid Barcode Primers at 10 μ M

Reagent	Volume
LongAmp Hot Start Taq 2X Master Mix	25 µl
Nuclease-free water	20 µl
Total volume	45 µl

Mix by pipetting, taking care not to generate bubbles or foam.

Note: The non-hot start product, LongAmp Taq 2X Master Mix (e.g. NEB, M0287) is also compatible with this method. If using this alternative, please ensure you follow the manufacturers instructions and prepare this reaction on ice.

5 Aliquot out the master mix into a new separate 0.2 ml thin-walled PCR tube for each sample preparation.

6 To each tube containing the prepared master mix, add the Rapid Barcode Primers (01-24). Each tube should have a different barcode:

Note: To make sample tracking easier, we recommend that the two types of sample preparation are given different blocks of barcodes:

e.g. Barcodes 1-4 for the viral samples and barcodes 5-8 for the bacterial/fungal samples.

For plate format with Rapid Barcode Primers at 1 µM

Reagent	Volume
Tube containing the prepared master mix (from previous step)	36 µl
Rapid Barcode Primers (01-24, at 1 µM)	10 µl
Total	46 µl

For vial format with Rapid Barcode Primers at 10 µM

Reagent	Volume
Tube containing the prepared master mix (from previous step)	45 µl
Rapid Barcode Primers (01-24, at 10 µM)	1 µl
Total	46 µl

7 To each tube containing master mix and a Rapid Barcode Primer, add your RT (viral) sample or tagmented (bacterial/fungal) sample:

For viral samples:

- Add 5 µl RT sample to the reaction tube. **Do not deviate from the recommended volume** as this can lead to sub-optimal results.

Reagent	Volume
Master mix + Rapid Barcode Primer	46 µl
RT viral sample	5 µl
Total	51 µl

Note: If you are concerned that you will not reach the minimum molarity, we recommend assembling two separate reactions for your viral sample using the volumes above, and pooling following PCR. **Do not** alter the PCR volumes as this will result in poor performance of the PCR and lead to sub-optimal results.

For bacterial/fungal samples:

- Add the full volume of tagmented sample to the reaction tube:

Reagent	Volume
Master mix + Rapid Barcode Primer	46 µl
Tagmented bacterial/fungal DNA sample	~4 µl
Total	50 µl

8 Mix gently by flicking the tube, and spin down.

9 Amplify using the following cycling conditions:

Cycle step	Temperature	Time	No. of cycles
Initial denaturation	95°C	45 seconds	1
Denaturation	95°C	15 seconds	30
Annealing	56°C	15 seconds	
Extension	65°C	4 minutes	
Final extension	65°C	6 minutes	1
Hold	10°C	∞	

Note: Total PCR time is 2 hours 43 minutes. Timing may differ depending on equipment and ambient conditions.

Optional: The PCR amplification can be performed and left at the hold temperature overnight.



Take forward your sample into the clean up, quantification and adapter attachment step.

5. Clean-up, quantification and adapter attachment

Materials

AMPure XP Beads (AXP)
Elution Buffer (EB)
Rapid Adapter (RA)
Adapter Buffer (ADB)

Consumables

Thermolabile Exonuclease I (NEB, M0568)
Freshly prepared 80% ethanol in nuclease-free water
1.5 ml Eppendorf DNA LoBind tubes
0.2 ml thin-walled PCR tubes
Qubit™ dsDNA HS Assay Kit (ThermoFisher, Q32851)

Equipment	Thermal cycler
	Hula mixer (gentle rotator mixer)
	Magnetic separation rack
	Qubit fluorometer (or equivalent)
	Timer
	P1000 pipette and tips
	P200 pipette and tips
	P100 pipette and tips
	P20 pipette and tips
	P10 pipette and tips
	P2 pipette and tips



Check your flow cell.

We recommend performing a flow cell check before completing your library prep to ensure you have a flow cell with enough pores for a good sequencing run.

See the [flow cell check document](#) for more information.

1 Quantify the sample tubes (PCR products from the previous step) using the Qubit dsDNA HS Assay Kit.

Make a note of each samples concentration.

Note: Sample concentration may vary depending on input. Some of your samples might be below the limit of detection for the Qubit dsDNA HS Assay Kit. This is not an uncommon occurrence.

Please proceed with all of your samples, following the recommendations in the instructions below.

2 Add 1 µl of Thermolabile Exo I to each of the sample tubes (PCR product) and mix by pipetting.

3 Incubate the reactions using the following conditions:

Temperature	Time
37°C	10 minutes
80°C	1 minute
10°C	Hold

- 4 In a new 1.5 ml Eppendorf DNA LoBind tube, pool all barcoded samples in equimolar ratios to a combined final concentration of 800 ng.**

For example, if 10 barcodes were used, take forward 80 ng of each sample.

Note: In cases where the concentration of your sample PCR product is too low, take forward the maximum available volume.

- 5 Resuspend the AMPure XP beads (AXP) by vortexing.**
- 6 Add 0.6X volume of AMPure XP beads (AXP) to the pooled samples.**

Ensure you have accurately measured the volume of your pooled samples to maintain the 0.6X ratio of the AMPure XP beads (AXP).
- 7 Incubate on a Hula mixer (rotator mixer) for 5 minutes at room temperature.**
- 8 Prepare at least 1 ml of fresh 80% ethanol in nuclease-free water.**
- 9 Briefly spin down the sample and pellet on a magnetic rack until supernatant is clear and colourless. Keep the tube on the magnetic rack, and pipette off the supernatant.**
- 10 Keep the tube on the magnet and wash the beads with 500 µl of freshly-prepared 80% ethanol without disturbing the pellet. Remove the ethanol using a pipette and discard.**

- 11 Repeat the previous step.**

- 12** Spin down and place the tube back on the magnet. Pipette off any residual ethanol. Allow to dry for ~30 seconds, but do not dry the pellet to the point of cracking.
- 13** Remove the tube from the magnetic rack and resuspend the pellet in 15 µl Elution Buffer (EB).
- 14** Incubate for 2 minutes at room temperature.
- 15** Pellet the beads on a magnet until the eluate is clear and colourless.
- 16** Remove and retain 14 µl of eluate containing the DNA library into a clean 1.5 ml Eppendorf DNA LoBind tube.
 - Remove and retain the eluate which contains the DNA library in a clean 1.5 ml Eppendorf DNA LoBind tube.
 - Dispose of the pelleted beads.



Quantify 1 µl of eluted sample using a Qubit fluorometer and the Qubit dsDNA HS Assay Kit.

Expected yield of 10-40 ng/µl

- 17** Take forward 11 µl of your eluted samples into a clean 1.5 ml Eppendorf DNA LoBind tube.
- 18** In a fresh 1.5 ml Eppendorf DNA LoBind tube, dilute the Rapid Adapter (RA) as follows and pipette mix:

Reagent	Volume
Rapid Adapter (RA)	1.5 µl
Adapter Buffer (ADB)	3.5 µl
Total	5 µl

19 Add 1 µl of the diluted Rapid Adapter (RA) to the barcoded DNA.

20 Mix gently by flicking the tube, and spin down.

21 Incubate the reaction for 5 minutes at room temperature.



The prepared library is used for loading into the flow cell. Store the library on ice or at 4°C until ready to load.

6. Priming and loading the MinION and GridION Flow Cell

Materials

Flow Cell Flush (FCF)
Flow Cell Tether (FCT)
Library Beads (LIB)
Sequencing Buffer (SB)

Consumables

MinION/GridION Flow Cell
Bovine Serum Albumin (BSA) (50 mg/ml) (e.g. Invitrogen™ UltraPure™ BSA 50 mg/ml, AM2616)
1.5 ml Eppendorf DNA LoBind tubes

Equipment

MinION or GridION device
MinION/GridION Flow Cell Light Shield
P1000 pipette and tips
P100 pipette and tips
P20 pipette and tips
P10 pipette and tips



Please note, this kit is only compatible with R10.4.1 flow cells (FLO-MIN114).



Take the flow cell out of the fridge and leave it at room temperature for 20 minutes. This will improve visibility of the array during priming and sample loading.



Priming and loading a flow cell

We recommend all new users watch the '[Priming and loading your flow cell](#)' video before your first run.

- 1 **Thaw the Sequencing Buffer (SB), Library Beads (LIB), Flow Cell Tether (FCT) and Flow Cell Flush (FCF) at room temperature before mixing by vortexing. Then spin down and store on ice.**



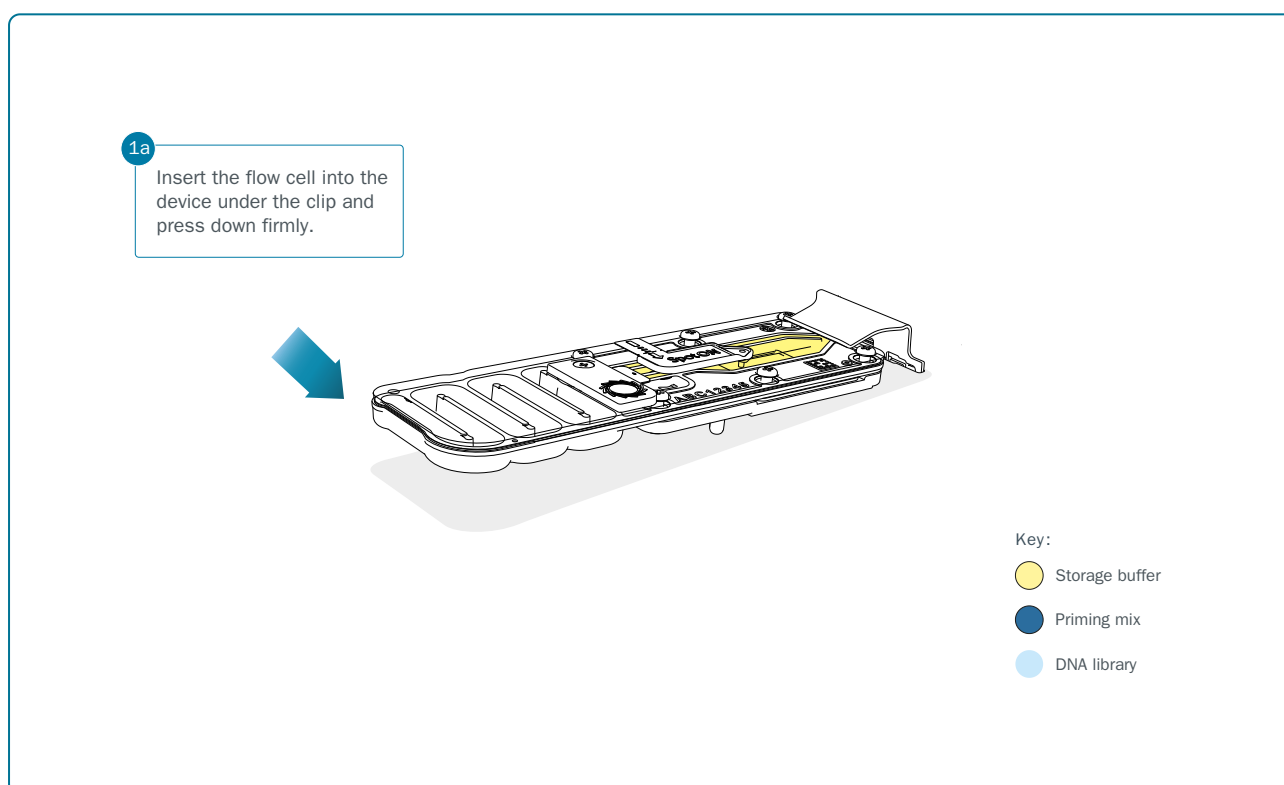
For optimal sequencing performance and improved output on MinION R10.4.1 flow cells (FLO-MIN114), add Bovine Serum Albumin (BSA) to the flow cell priming mix at a final concentration of 0.2 mg/ml.

Note: We do not recommend using any other albumin type (e.g. recombinant human serum albumin).

- 2 **To prepare the flow cell priming mix with BSA, combine the following reagents in a fresh 1.5 ml Eppendorf DNA LoBind tube. Mix by inverting the tube and pipette mix at room temperature:**

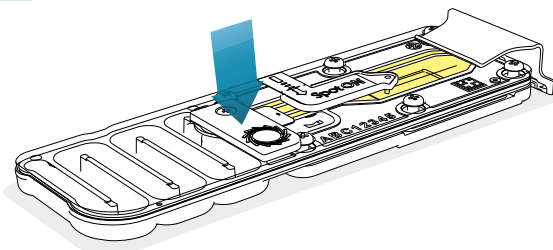
Reagents	Volume per flow cell
Flow Cell Flush (FCF)	1,170 µl
Bovine Serum Albumin (BSA) at 50 mg/ml	5 µl
Flow Cell Tether (FCT)	30 µl
Final total volume in tube	1,205 µl

- 3 Open the MinION or GridION device lid and slide the flow cell under the clip. Press down firmly on the priming port cover to ensure correct thermal and electrical contact.



1b

Insert the flow cell into the device under the clip and press down firmly on the Priming port cover.



Complete a flow cell check to assess the number of pores available before loading the library.

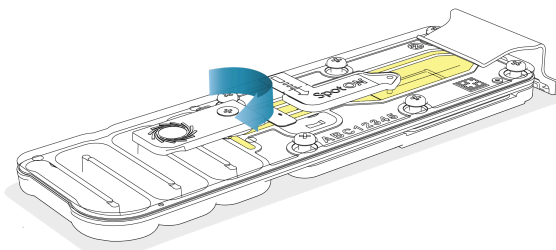
This step can be omitted if the flow cell has been checked previously.

See the [flow cell check document](#) for more information.

4

Slide the flow cell priming port cover clockwise to open the priming port.

2 Slide open the **Priming port** cover.



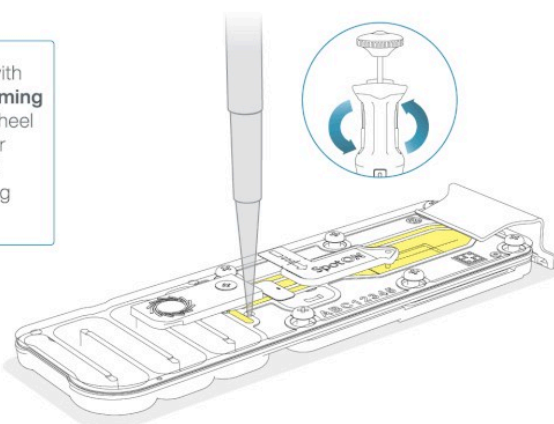
Take care when drawing back buffer from the flow cell. Do not remove more than 20-30 μ l, and make sure that the array of pores are covered by buffer at all times. Introducing air bubbles into the array can irreversibly damage pores.

5 After opening the priming port, check for a small air bubble under the cover. Draw back a small volume to remove any bubbles:

1. Set a P1000 pipette to 200 μ l
2. Insert the tip into the priming port
3. Turn the wheel until the dial shows 220-230 μ l, to draw back 20-30 μ l, or until you can see a small volume of buffer entering the pipette tip

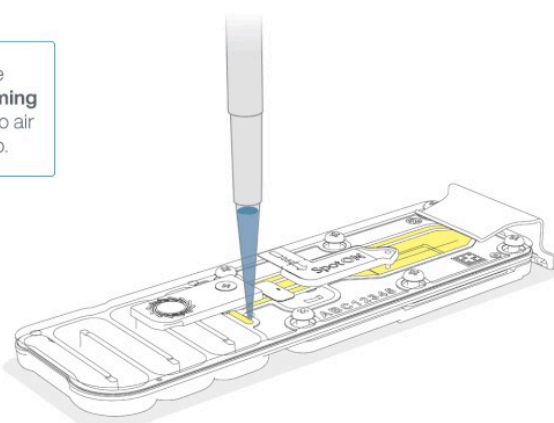
Note: Visually check that there is continuous buffer from the priming port across the sensor array.

- 3** Insert a P1000 pipette with an empty tip into the **Priming port**. Turn the pipette wheel to draw back 20-30 µl or until you can see a small volume of buffer entering the pipette tip.



- 6** Load 800 µl of the priming mix into the flow cell via the priming port, avoiding the introduction of air bubbles. Wait for five minutes. During this time, prepare the library for loading by following the steps below.

- 4** Slowly load 800 µl of the priming mix into the **Priming port**. Ensure there are no air bubbles in the pipette tip.



Wait 5 minutes before proceeding to the next step.

- 7** Thoroughly mix the contents of the Library Beads (LIB) by pipetting.



The Library Beads (LIB) tube contains a suspension of beads. These beads settle very quickly. It is vital that they are mixed immediately before use.

We recommend using the Library Beads (LIB) for most sequencing experiments. However, the Library Solution (LIS) is available for more viscous libraries.

- 8 In a new 1.5 ml Eppendorf DNA LoBind tube, prepare the library for loading as follows:**

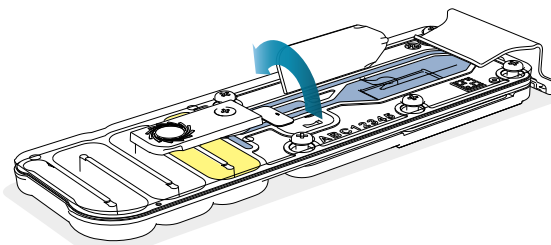
Reagent	Volume per flow cell
Sequencing Buffer (SB)	37.5 µl
Library Beads (LIB) mixed immediately before use	25.5 µl
DNA library	12 µl
Total	75 µl

- 9 Complete the flow cell priming:**

1. Gently lift the SpotON sample port cover to make the SpotON sample port accessible.
2. Load **200 µl** of the priming mix into the flow cell priming port (**not** the SpotON sample port), avoiding the introduction of air bubbles.

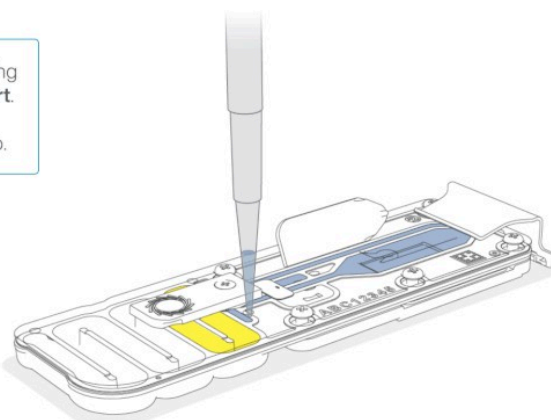
5

Gently flip open the SpotON sample port cover.



6

Load 200 μ l of the priming mix into the **Priming Port**. Ensure there are no air bubbles in the pipette tip.



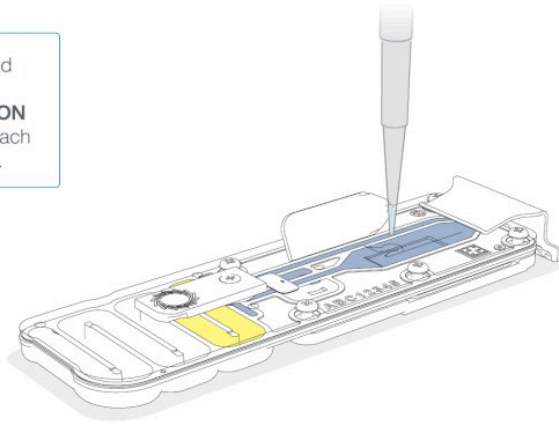
10 Mix the prepared library gently by pipetting up and down just prior to loading.

11 Add 75 μ l of the prepared library to the flow cell via the SpotON sample port in a dropwise fashion. Ensure each drop flows into the port before adding

the next.

7

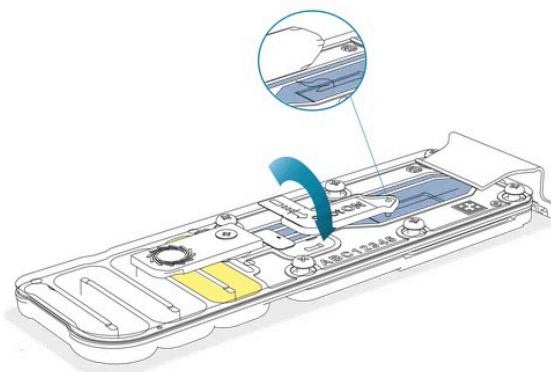
Pipette mix the prepared library and load 75 µl dropwise into the **SpotON** sample port, ensuring each drop flows into the port.



- 12 Gently replace the SpotON sample port cover, making sure the bung enters the SpotON port and close the priming port.

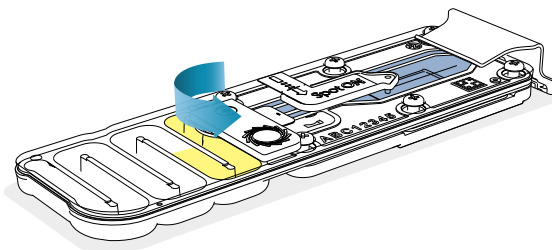
8

Gently replace the **SpotON** sample port cover.



9

Gently close the
Priming port.



For optimal sequencing output, install the light shield on your flow cell as soon as the library has been loaded.

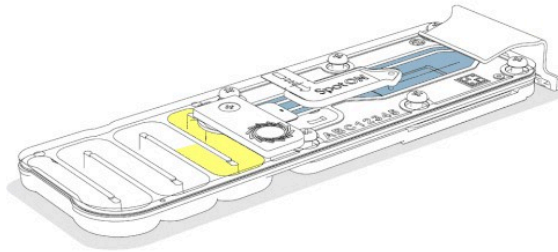
We recommend leaving the light shield on the flow cell when library is loaded, including during any washing and reloading steps. The shield can be removed when the library has been removed from the flow cell.

13 Place the light shield onto the flow cell, as follows:

1. Carefully place the leading edge of the light shield against the clip. **Note:** Do not force the light shield underneath the clip.
2. Gently lower the light shield onto the flow cell. The light shield should sit around the SpotON cover, covering the entire top section of the flow cell.

10

Carefully align the **light shield** against the clip and lower onto the flow cell.



The MinION Flow Cell Light Shield is not secured to the flow cell and careful handling is required after installation.



Close the device lid and set up a sequencing run on MinKNOW.



When a flow cell is inserted into the MinION Mk1D, the device lid will sit on top of the flow cell, leaving a small gap around the sides. This is normal and has no impact on the performance of the device.

Please refer to this [FAQ](#) regarding the device lid.



7. Data acquisition and basecalling



We do not recommend sequencing and performing data analysis simultaneously on your device.

To ensure the compute on your device can keep up with the requirements for sequencing and/or analysis, **we strongly recommend against running both processes at the same time.**

Ensure your sequencing run has completed before setting off data analysis. Data analysis will be performed post-sequencing.

Equally, we do not recommend starting a sequencing run if you are currently performing data analysis on your device.

How to start sequencing

The MinKNOW software controls the sequencing device and carries out data acquisition and real-time basecalling. Please ensure MinKNOW is installed on your computer or device. Further instructions for setting up a sequencing run can be found in the [MinKNOW protocol](#).

We recommend setting up your sequencing run using the basecalling and barcoding recommendations outlined below. All other parameters can be left to their default settings.

MinKNOW settings for rapid metagenomic sequencing

For the fastest turnaround time and easiest analysis, basecalling should be performed live during the sequencing run using the High Accuracy (HAC) basecaller.

Below are the recommendations for MinKNOW settings:

Positions

Flow cell position: [user defined]

Experiment name: [user defined]

Flow cell type: FLO-MIN114

Sample ID: [user defined]

Kit

Kit selection: Rapid PCR Barcoding Kit (SQK-RPB114-24)

Run configuration

Sequencing and analysis

Basecalling: On [default]

Modified bases: Off

Model: High-accuracy basecalling (HAC) [default]

Barcoding: On [default]

Trim barcodes: Off [default]

Barcode both ends: Off [default]

Custom barcodes selection: Off [default]

Alignment: Off [default]

Adaptive sampling: Off [default]

Advanced options

Active channel selection: On [default]

Time between pore scans: 1.5 [default]

Reserve pores: On [default]

Data targets

Run limit: [user defined]*

*Sequencing time will depend on data requirements. For rapid information, data can be analysed after as little as 30 minutes of sequencing. Or to maximise data generation, you can sequence for up to 72 hours.

Output

Output format

.POD5: On [default]

.FASTQ: On [default]

.BAM: On

Filtering: On [default]

Qscore: 9 [default]

Minimum read length: 200 bp [default]

8. Downstream analysis



We do not recommend sequencing and performing data analysis simultaneously on your device.

To ensure the compute on your device can keep up with the requirements for sequencing and/or analysis, **we strongly recommend against running both processes at the same time.**

Ensure your sequencing run has completed before setting off data analysis. Data analysis will be performed post-sequencing.

Equally, we do not recommend starting a sequencing run if you are currently performing data analysis on your device.

Post-basecalling analysis

We recommend performing downstream analysis using [EPI2ME](#) which facilitates bioinformatic analyses by allowing users to run Nextflow workflows in a desktop application. EPI2ME maintains a collection of bioinformatic workflows which are curated and actively maintained by experts in long-read sequence analysis.

Follow the instructions in the [EPI2ME Installation guide](#) to install the application on your device. For more information on how to use EPI2ME, refer to the [EPI2ME Quick Start guide](#).

The basecalled data generated by the sequencing software can be easily analysed using the [wf-metagenomics workflow](#), a bioinformatic pipeline written in Nextflow. This workflow provides identification and abundance estimation of taxa present in your sample.

Using the recommended database, human, fungal, bacterial, viral, archaea and protozoal sequences can be accurately identified using the workflow.

We recommend you always use the latest available version of the workflow. Further information about the usage and results provided by the workflow can be found in the [wf-metagenomics EPI2ME documentation page](#).

Notes

- You can also run this workflow through command line. However, we only recommend this option for experienced users. For more information and the latest version of the workflow, please visit the [wf-metagenomics page on GitHub](#). If using the workflow through command line, the settings recommended for optimal workflow performance for this protocol are: `--kraken2_confidence 0.01 --database_set PlusPF-8 --store_dir /path/to/database/download/directory/`
- `/path/to/database/download/directory/` is a placeholder. The exact location used will depend on the user system.



Test data is available on Github to test the wf-metagenomics workflow.

The test data can be found in the following repository: [wf-metagenomics test data](#).

Running wf-metagenomics with EPI2ME

Within the EPI2ME app, new workflows can be installed by selecting the appropriate workflow under the "Available Workflows" tab in the Workflows section.

Already installed workflows can be found under the "Installed" tab.

You can test your installation using a small test dataset that is provided with the workflow:

1. Launch the EPI2ME app.
2. Select **View workflows**.
3. Select the **wf-metagenomics** workflow.
4. Select **Use demo data**.
5. Launch the workflow.
6. Once the run has completed, results can be viewed under the "Report" tab. The workflow outputs an interactive HTML report.

Running the workflow

To run wf-metagenomics with the settings recommended for the dual arm metagenomics protocol:

1. Launch the EPI2ME app.
2. Select **View workflows**.
3. Select the **wf-metagenomics** workflow.
4. Select **Run this workflow**.
5. Choose local or cloud option (if available).
6. In "Input Options", specify the input data for the workflow (either FASTQ or BAM).
7. In "Sample Options", provide a sample sheet in comma-separated values (CSV) format. We suggest you provide an alias for each barcode that describes both the sample and prep type (e.g. sample_A_viral, sample_A_bacterial).
8. In "Reference Options", choose **PlusPF-8 database** to ensure archaea, bacterial, viral, human, and fungal taxa can be identified by the workflow.
9. In "Kraken2 Options", set **Confidence score threshold** to **0.01**.
10. Click **Launch workflow**.
11. Once the run has completed, results can be viewed under the "Report" tab. The workflow outputs an interactive HTML report.

Note: All options not specified should be left to their default values.

Optional: If you need to adjust the computational resource available to your workflow, you can change local CPU and memory allocation in the workflow setup.

9. Flow cell reuse and returns

Materials Flow Cell Wash Kit (EXP-WSH004)

- 1 **After your sequencing experiment is complete, if you would like to reuse the flow cell, please follow the Flow Cell Wash Kit protocol and store the washed flow cell at +2°C to +8°C.**

The [Flow Cell Wash Kit protocol](#) is available on the Nanopore Community.



We recommend you to wash the flow cell as soon as possible after you stop the run. However, if this is not possible, leave the flow cell on the device and wash it the next day.

2 Alternatively, follow the returns procedure to send the flow cell back to Oxford Nanopore.

Instructions for returning flow cells can be found [here](#).



If you encounter issues or have questions about your sequencing experiment, please refer to the Troubleshooting Guide that can be found in this protocol.

10. Issues during DNA/RNA extraction and library preparation

Below is a list of the most commonly encountered issues, with some suggested causes and solutions.

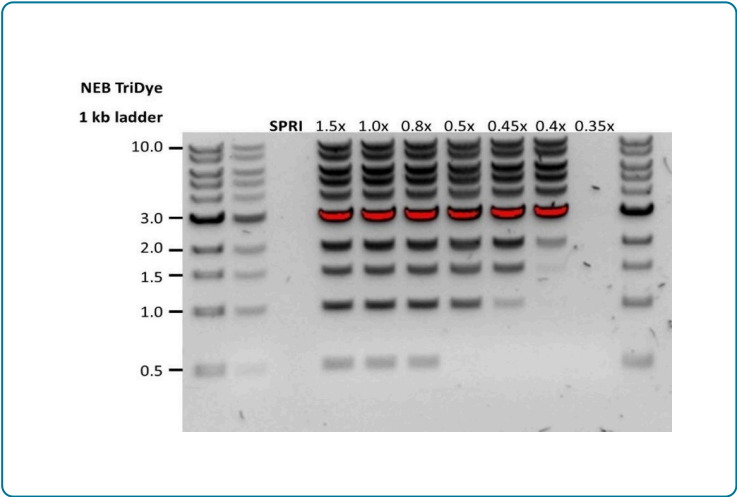
We also have an FAQ section available on the [Nanopore Community Support](#) section.

If you have tried our suggested solutions and the issue still persists, please contact Technical Support via email (support@nanoporetech.com) or via [LiveChat in the Nanopore Community](#).

Low sample quality

Observation	Possible cause	Comments and actions
Low DNA purity (Nanodrop reading for DNA OD 260/280 is <1.8 and OD 260/230 is <2.0–2.2)	The DNA extraction method does not provide the required purity	The effects of contaminants are shown in the Contaminants document. Please try an alternative extraction method that does not result in contaminant carryover. Consider performing an additional SPRI clean-up step.
Low RNA integrity (RNA integrity number <9.5 RIN, or the rRNA band is shown as a smear on the gel)	The RNA degraded during extraction	Try a different RNA extraction method. For more info on RIN, please see the RNA Integrity Number document. Further information can be found in the DNA/RNA Handling page.
RNA has a shorter than expected fragment length	The RNA degraded during extraction	Try a different RNA extraction method. For more info on RIN, please see the RNA Integrity Number document. Further information can be found in the DNA/RNA Handling page. We recommend working in an RNase-free environment, and to keep your lab equipment RNase-free when working with RNA.

Low DNA recovery after AMPure bead clean-up

Observation	Possible cause	Comments and actions
Low recovery	DNA loss due to a lower than intended AMPure beads-to-sample ratio	1. AMPure beads settle quickly, so ensure they are well resuspended before adding them to the sample. 2. When the AMPure beads-to-sample ratio is lower than 0.4:1, DNA fragments of any size will be lost during the clean-up.
Low recovery	DNA fragments are shorter than expected	The lower the AMPure beads-to-sample ratio, the more stringent the selection against short fragments. Please always determine the input DNA length on an agarose gel (or other gel electrophoresis methods) and then calculate the appropriate amount of AMPure beads to use.
		
Low recovery after end-prep	The wash step used ethanol <70%	DNA will be eluted from the beads when using ethanol <70%. Make sure to use the correct percentage.

11. Issues during the sequencing run using a Rapid-based sequencing kit

Below is a list of the most commonly encountered issues, with some suggested

causes and solutions.

We also have an FAQ section available on the [Nanopore Community Support](#) section.

If you have tried our suggested solutions and the issue still persists, please contact Technical Support via email (support@nanoporetech.com) or via [LiveChat in the Nanopore Community](#).

Fewer pores at the start of sequencing than after Flow Cell Check

Observation	Possible cause	Comments and actions
MinKNOW reported a lower number of pores at the start of sequencing than the number reported by the Flow Cell Check	An air bubble was introduced into the nanopore array	After the Flow Cell Check it is essential to remove any air bubbles near the priming port before priming the flow cell. If not removed, the air bubble can travel to the nanopore array and irreversibly damage the nanopores that have been exposed to air. The best practice to prevent this from happening is demonstrated in this video .
MinKNOW reported a lower number of pores at the start of sequencing than the number reported by the Flow Cell Check	The flow cell is not correctly inserted into the device	Stop the sequencing run, remove the flow cell from the sequencing device and insert it again, checking that the flow cell is firmly seated in the device and that it has reached the target temperature. If applicable, try a different position on the device (GridION/PromethION).
MinKNOW reported a lower number of pores at the start of sequencing than the number reported by the Flow Cell Check	Contaminations in the library damaged or blocked the pores	The pore count during the Flow Cell Check is performed using the QC DNA molecules present in the flow cell storage buffer. At the start of sequencing, the library itself is used to estimate the number of active pores. Because of this, variability of about 10% in the number of pores is expected. A significantly lower pore count reported at the start of sequencing can be due to contaminants in the library that have damaged the membranes or blocked the pores. Alternative DNA/RNA extraction or purification methods may be needed to improve the purity of the input material. The effects of contaminants are shown in the Contaminants Know-how piece . Please try an alternative extraction method that does not result in contaminant carryover.

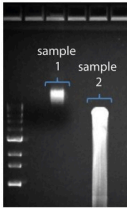
MinKNOW script failed

Observation	Possible cause	Comments and actions
MinKNOW shows "Script failed"		Restart the computer and then restart MinKNOW. If the issue persists, please collect the MinKNOW log files and contact Technical Support. If you do not have another sequencing device available, we recommend storing the flow cell and the loaded library at 4°C and contact Technical Support for further storage guidance.

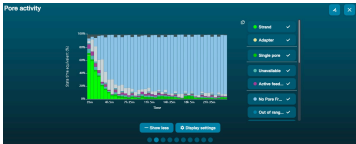
Pore occupancy below 40%

Observation	Possible cause	Comments and actions
Pore occupancy <40%	Not enough library was loaded on the flow cell	Ensure the correct concentration of good quality library is loaded on to a MinION/GridION flow cell. To check the concentration, please refer to the library preparation protocol. Please quantify the library before loading and calculate mols using tools like the Promega Biomath Calculator , choosing "dsDNA: µg to pmol"
Pore occupancy close to 0	The Rapid Sequencing Kit V14/Rapid Barcoding Kit V14 was used, and sequencing adapters did not attach to the DNA	Make sure to closely follow the protocol and use the correct volumes and incubation temperatures. A Lambda control library can be prepared to test the integrity of reagents.
Pore occupancy close to 0	No tether on the flow cell	Tethers are adding during flow cell priming (FCT tube). Make sure FCT was added to FCF before priming.

Shorter than expected read length

Observation	Possible cause	Comments and actions
Shorter than expected read length	Unwanted fragmentation of DNA sample	Read length reflects input DNA fragment length. Input DNA can be fragmented during extraction and library prep. 1. Please review the Extraction Methods in the Nanopore Community for best practice for extraction. 2. Visualise the input DNA fragment length distribution on an agarose gel before proceeding to the library prep.  In the image above, Sample 1 is of high molecular weight, whereas Sample 2 has been fragmented. 3. During library prep, avoid pipetting and vortexing when mixing reagents. Flicking or inverting the tube is sufficient.

Large proportion of unavailable pores

Observation	Possible cause	Comments and actions
Large proportion of unavailable pores (shown as blue in the channels panel and pore activity plot)  The pore activity plot above shows an increasing proportion of "unavailable" pores over time.	Contaminants are present in the sample	Some contaminants can be cleared from the pores by the unblocking function built into MinKNOW. If this is successful, the pore status will change to "sequencing pore". If the portion of unavailable pores stays large or increases: 1. A nuclease flush using the Flow Cell Wash Kit (EXP-WSH004) can be performed, or 2. Run several cycles of PCR to try and dilute any contaminants that may be causing problems.

Large proportion of inactive pores

Observation	Possible cause	Comments and actions
Large proportion of inactive/unavailable pores (shown as light blue in the channels panel and pore activity plot. Pores or membranes are irreversibly damaged)	Air bubbles have been introduced into the flow cell	Air bubbles introduced through flow cell priming and library loading can irreversibly damage the pores. Watch the Priming and loading your flow cell video for best practice
Large proportion of inactive/unavailable pores	Certain compounds co-purified with DNA	Known compounds, include polysaccharides, typically associate with plant genomic DNA. 1. Please refer to the Plant leaf DNA extraction method. 2. Clean-up using the QIAGEN PowerClean Pro kit. 3. Perform a whole genome amplification with the original gDNA sample using the QIAGEN REPLI-g kit.
Large proportion of inactive/unavailable pores	Contaminants are present in the sample	The effects of contaminants are shown in the Contaminants Know-how piece. Please try an alternative extraction method that does not result in contaminant carryover.

Temperature fluctuation

Observation	Possible cause	Comments and actions
Temperature fluctuation	The flow cell has lost contact with the device	Check that there is a heat pad covering the metal plate on the back of the flow cell. Re-insert the flow cell and press it down to make sure the connector pins are firmly in contact with the device. If the problem persists, please contact Technical Services.

Failed to reach target temperature

Observation	Possible cause	Comments and actions
MinKNOW shows "Failed to reach target temperature"	The instrument was placed in a location that is colder than normal room temperature, or a location with poor ventilation (which leads to the flow cells overheating)	MinKNOW has a default timeframe for the flow cell to reach the target temperature. Once the timeframe is exceeded, an error message will appear and the sequencing experiment will continue. However, sequencing at an incorrect temperature may lead to a decrease in throughput and lower q-scores. Please adjust the location of the sequencing device to ensure that it is placed at room temperature with good ventilation, then re-start the process in MinKNOW. Please refer to this link for more information on MinION temperature control.